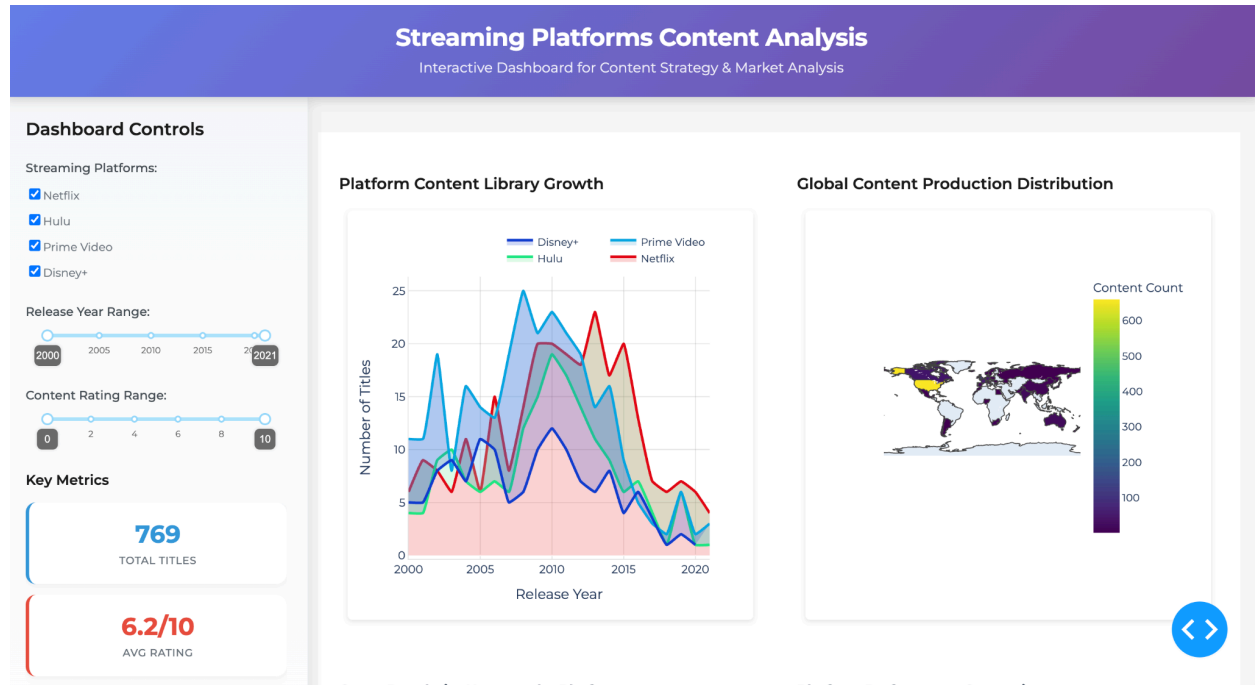


DATANVI Final Project Report

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GOAL VISUALIZATION: Streaming Platforms Content Analysis Dashboard

Target Audience

The primary users of this dashboard are content strategy teams and executives at Netflix, Hulu, Prime Video, and Disney+. These professionals make high-stakes decisions on content acquisition and production and are familiar with basic data visualization. They need fast, interactive tools for competitive analysis, spotting market gaps, and strategic planning. Secondary users include content creators and producers who want insights into platform preferences and potential market opportunities.

Problem Definition

Content strategy teams need to compare streaming platforms, but current tools like Excel can't handle real-time, multi-dimensional filtering. This makes it hard to explore trends across platforms, time, genres, and regions. The main problem is the lack of an interactive way to analyze content patterns, platform focus areas, seasonal trends, and production locations to guide decisions on what content to acquire, produce, or where to expand.

Dataset Descriptions and Metadata

We used two datasets: one with 16,744 titles and platform availability (Netflix, Hulu, Prime Video, Disney+), and another from TMDB with details like ratings, budgets, production countries, and vote counts. After cleaning and merging based on title, we kept 988 complete records from 1922 to 2023.

Missing Rotten Tomatoes data was dropped, and missing budgets were filled with the median. Key variables used include platform availability, release year, vote average, vote count, genres, production countries, budget, revenue, and release date.

Data Merging and Preparation Steps

We merged two datasets using an inner join based on matching titles, which gave us 988 common records. For the production_countries and spoken_languages columns, we used custom functions to extract data from the JSON fields, with error handling to avoid crashes. To make geographic analysis possible, we created a new countries_df by duplicating rows for each production country. We also added new features like content_age (calculated as current year minus release year) and release_month (from parsing the release date). Missing values in the budget column were filled using the median. Finally, we removed extra columns and cleaned up the column names so the dashboard could use them easily.

Visualizations

Chart 1: Platform Content Growth

Field	Description
Type	Area Chart (multi-series)
Data	X: Year, Y: Count, Color: Platform
Channels	Transparent overlays, smooth curves
Task	Reveal platform growth trends over time

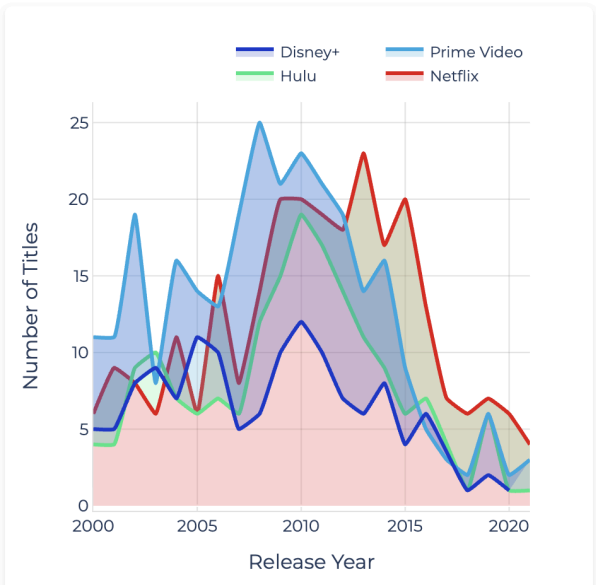


Chart 2: World Choropleth Map

Field	Description
Type	Choropleth Map
Data	Countries (ISO), content count
Channels	Color intensity, hover tooltip
Task	Identify production hotspots & market opportunities

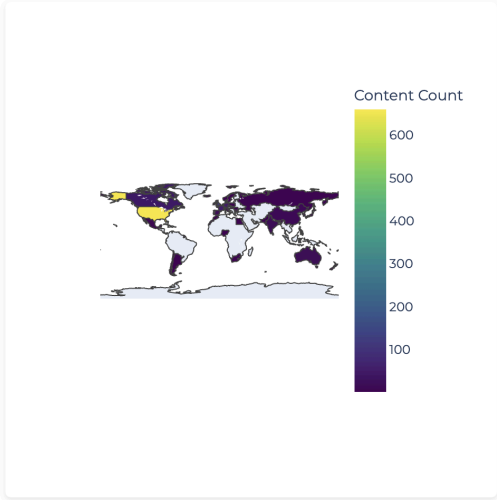


Chart 3: Genre-Platform Heatmap

Field	Description
Type	Heatmap
Data	X: Platforms, Y: Genres, Color: Count
Channels	Color matrix, interactive cells
Task	Analyze genre specialization & gaps

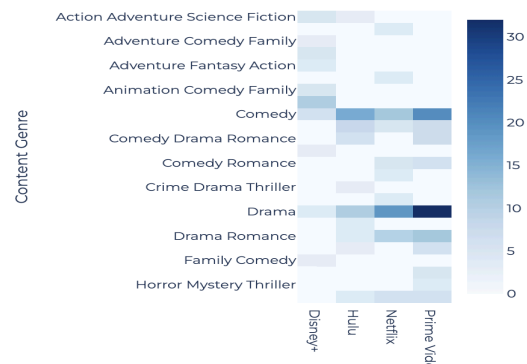


Chart 4: Platform Comparison Dashboard

Field	Description
Type	Violin + Bar Charts
Data	Ratings, content volume, vote counts
Channels	Statistical overlays, color-coded bars
Task	Benchmark platforms quantitatively

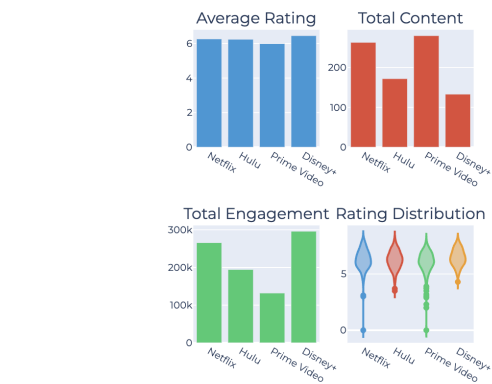


Chart 5: Seasonal Release Patterns

Field	Description
Type	Multi-line Chart
Data	X: Month, Y: Content Count
Channels	Colored lines, markers
Task	Discover release seasonality & timing patterns

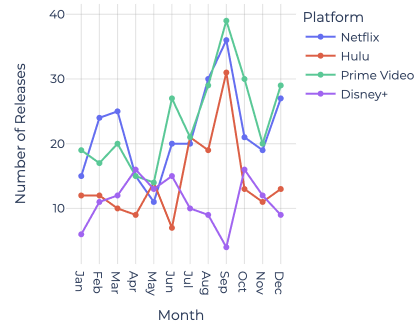


Chart 6: Ratings vs Engagement Correlation

Field	Description
Type	Scatter Plot
Data	X: Rating, Y: Vote Count
Channels	Point opacity, trend line
Task	Explore quality vs popularity trends

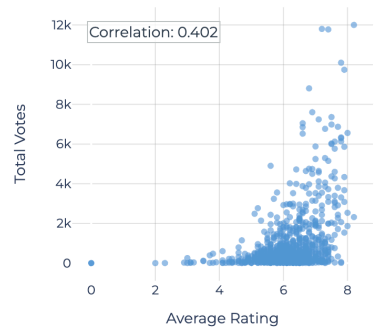
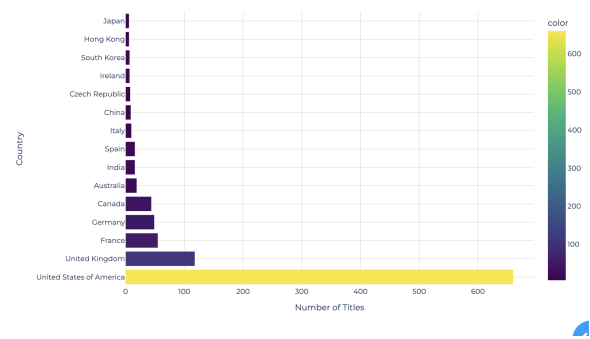


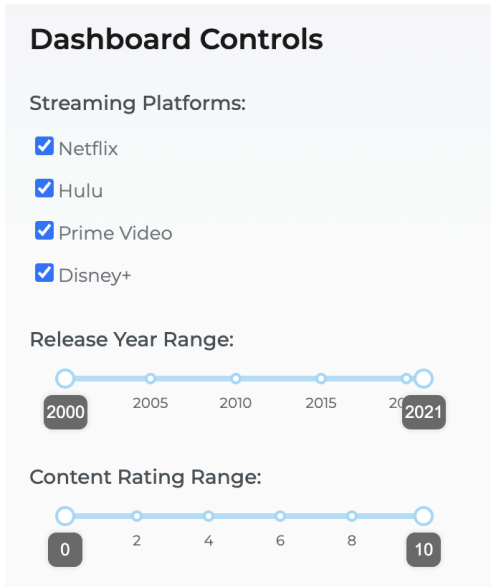
Chart 7: Top Content Producing Countries by Platform

Field	Description
Type	Horizontal Bar Chart (Grouped/Stacked)
Data	X: Countries, Y: Content Count, Groups: Platform
Channels	Color (by platform), height (content count)
Task	Compare content production volume across countries and platforms



Interactivity Design and Justification

The dashboard lets users filter all seven charts at once using platform checkboxes, a time range slider, and a rating threshold slider. Platform filtering uses boolean indexing with `df[selected_platforms].sum(axis=1) > 0` to include titles from any selected platform. Charts are connected—map data links to country rankings, and time data connects trends to current stats. If no platforms are selected or filters return nothing, the dashboard shows a message instead of breaking. This setup makes it easy to explore different patterns that static dashboards can’t handle.



For instance, a content strategist can instantly test scenarios like "Show me Disney+ comedy content from 2018–2020 in European markets" without the need for manual data reprocessing. The result is a more efficient and insightful way to analyze content trends and performance.